

DumpLens Design Document

AI-Assisted Multi-Source Phone Dump & Social Media Message Reconstruction Tool

Working product name: DumpLens

Alternative names: MessageIntel, CommsGraph, ThreadSift, CaseGraph Comms Module

Document type: Product and software design document

Primary users: Detectives, investigators, analysts, supervisors, prosecutors, task force personnel

Core principle: DumpLens helps investigators organize, compare, and review communication evidence. It does not replace investigator judgment, establish probable cause by itself, or label people as criminals without human review.

1. Product Summary

DumpLens is an offline-first investigative analysis tool designed to ingest multiple phone dumps, message exports, call logs, and social media message returns, then reconstruct communications across devices and platforms.

The tool is built around one major investigative problem:

In real cases, one device may be missing messages that still exist on another device, in a provider return, in screenshots, or inside a different app. Investigators need a fast, defensible way to compare sources, identify gaps, build timelines, and determine where more digging is needed.

DumpLens provides a structured workspace where users can:

- Import phone extraction message exports.
- Import social media message returns.
- Import call logs and related communication artifacts.
- Normalize inconsistent data into one common evidence model.
- Reconstruct conversations across devices and platforms.
- Identify messages present on one source but missing from another.
- Detect likely deletion gaps or incomplete extractions.
- Build unified timelines across SMS, iMessage, Instagram, Facebook, Snapchat, TikTok, WhatsApp, Signal, calls, and manual entries.
- Summarize priority conversations using AI or local LLMs.
- Extract names, nicknames, phone numbers, handles, vehicles, weapons terms, locations, and coded language.
- Suggest investigative gaps and potential warrant targets with source-backed reasoning.
- Export court/discovery-friendly reports with citations back to original artifacts.

DumpLens should feel like a practical investigator tool, not a generic chatbot.

2. Product Goals

2.1 Primary Goals

1. Fast triage of large message sets

Help investigators quickly understand hundreds of thousands of messages without manually reading every thread first.

2. Cross-device conversation reconstruction

Compare multiple dumps and message exports to reconstruct what likely occurred across all available sources.

3. Missing-message and deletion-gap detection

Highlight messages or time windows that appear on one source but not another.

4. Unified investigative timeline

Merge messages, calls, attachments, social media DMs, manual notes, and key case events into a single timeline.

5. Evidence-backed AI assistance

Every AI finding must cite the exact source message, timestamp, sender, recipient, device, platform, and import file.

6. Lead and warrant target generation

Suggest possible follow-up investigative leads and warrant targets based on source-backed gaps, unknown accounts, platform switches, or suspicious communication changes.

7. Human review and auditability

Allow investigators to approve, reject, edit, and document findings. Maintain a full audit trail.

2.2 Non-Goals

DumpLens should not:

- Automatically determine guilt.
- Automatically label a person as a gang member, suspect, co-conspirator, or criminal associate.
- Generate unsupported probable cause statements.
- Present AI interpretations as facts.
- Modify original evidence files.
- Hide uncertainty.
- Replace forensic extraction tools.
- Replace human review of source evidence.

3. Target Users

3.1 Primary User: Investigator

The investigator needs to:

- Import several dumps quickly.
- Search messages.
- Compare conversations.
- Find missing/deleted messages.
- Understand important threads.
- Build a timeline.
- Identify who or what to target next.
- Prepare notes, reports, warrant support, or case briefings.

3.2 Secondary User: Crime Analyst

The analyst needs to:

- Normalize large datasets.
- Identify patterns across multiple people.
- Build link charts and communication summaries.
- Prepare intelligence products for investigators.

3.3 Secondary User: Supervisor

The supervisor needs to:

- Review findings.
- Confirm that conclusions are source-backed.
- Track open leads.
- Ensure the case team is not relying on unsupported AI output.

3.4 Secondary User: Prosecutor

The prosecutor needs to:

- Review relevant messages in context.
 - Understand what source each finding came from.
 - See the distinction between artifact, inference, and investigator conclusion.
 - Export clean, defensible summaries.
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4. Core Use Cases

4.1 Homicide Communication Reconstruction

An investigator has:

- Suspect phone extraction.
- Victim phone extraction.
- Co-defendant phone extraction.
- Instagram provider return.
- Facebook messages.
- Call detail records.
- Screenshots from a witness.

DumpLens helps compare all sources, reconstruct the timeline before and after the homicide, identify deleted messages, and show which accounts or numbers need additional warrants.

4.2 Gang Retaliation Investigation

Investigators import:

- Multiple phone dumps.
- Social media exports.
- Jail call transcripts.
- Known gang slang dictionary.
- Prior incident timeline.

DumpLens highlights retaliation language, nickname connections, communication spikes, platform switching, and messages tied to specific incidents.

4.3 Missing Message Review

The victim's phone contains a message thread with the suspect. The suspect's phone has the same thread but is missing several messages during the offense window.

DumpLens flags:

- Which messages exist on the victim phone but not the suspect phone.
- Whether the missing messages are surrounded by normal matched traffic.
- Whether the missing window coincides with the incident.
- Whether the suspect later says things like "delete that," "call me," or "hit me on snap."

4.4 Warrant Planning

The tool identifies that:

- A suspect repeatedly tells others to move to Instagram.
- The suspect phone has no Instagram messages recovered.
- A co-defendant's dump contains Instagram DMs with the suspect during the offense window.
- A new unknown number appears immediately after the homicide.

DumpLens creates a lead packet suggesting possible warrant targets, with supporting message references.

5. Guiding Design Principles

5.1 Evidence First

The source artifact is always more important than the AI summary.

Every finding should answer:

- What is the claim?
- What source supports it?
- What is the exact message or artifact?
- What is fact versus inference?
- Who reviewed it?

5.2 Fast Triage, Slow Conclusion

DumpLens should help investigators find relevant material fast. It should not encourage rushed conclusions.

The UI should make it easy to:

- Mark something as interesting.
- Pin it to a timeline.
- Ask for more review.
- Reject weak AI findings.
- Add investigator notes.

5.3 Clear Confidence Language

All AI findings should use careful labels:

- Confirmed artifact
- Possible match

- Likely match
- Ambiguous match
- Possible deletion gap
- Needs investigator review
- Investigator-confirmed finding

Avoid labels like:

- Guilty
- Criminal associate
- Gang member
- Proven conspiracy
- Probable cause established

5.4 Source Transparency

The user should never wonder where a summary came from. Every summary, extracted entity, timeline event, or warrant suggestion should be expandable to show the supporting source rows.

5.5 Offline-First and Privacy-Conscious

The system should support local processing for sensitive evidence. Cloud AI should be optional, controlled, logged, and ideally redacted before use.

5.6 Investigator-Friendly Layout

The UI should not feel like an academic data science tool. It should feel like a practical case workspace with clear buttons, obvious filters, and plain-language labels.

6. High-Level Software Layers

DumpLens should be designed as layered software so each major function can evolve independently.

6.1 Presentation Layer

Responsible for:

- User interface.
- Workspaces.
- Dashboards.
- Review panels.
- Timeline views.
- Conversation views.
- Report preview.

Main technologies can vary, but a desktop-first approach is preferred.

Potential options:

- WPF / WinUI for Windows-native app.
- Avalonia for cross-platform desktop.
- Electron / Tauri for web-style UI.
- Blazor Hybrid if using .NET.
- React front end with local backend.

6.2 Application Layer

Responsible for:

- Case workflows.
- Import wizard logic.
- Review queue actions.
- Timeline pinning.
- Finding approval/rejection.
- Report generation orchestration.
- User permissions.

This layer coordinates services without directly owning raw data parsing or AI logic.

6.3 Ingestion Layer

Responsible for:

- Importing CSV, XLSX, JSON, XML, PDF/transcripts where feasible.
- Handling multiple export formats.
- Mapping source columns to DumpLens fields.
- Validating imported files.
- Hashing original files.
- Preserving import metadata.
- Creating normalized records.

6.4 Normalization Layer

Responsible for converting messy source data into common internal models:

- Person
- Identity
- Phone number
- Social media handle
- Device
- Source file
- Platform

- Message
- Call
- Attachment
- Conversation
- Timeline event
- Extracted entity
- Finding
- Lead

6.5 Reconciliation Layer

Responsible for comparing messages across sources:

- Match same messages across devices.
- Detect source-only messages.
- Detect missing counterpart messages.
- Identify possible deletion gaps.
- Resolve duplicate records.
- Cluster related conversations.
- Link messages to people, devices, and platforms.

6.6 AI / LLM Analysis Layer

Responsible for optional AI-assisted analysis:

- Thread summaries.
- Entity extraction.
- Slang/code term review.
- Timeline event suggestions.
- Communication pattern changes.
- Platform-switching detection.
- Lead suggestions.
- Warrant target suggestions.

The AI layer must return structured output with source references, not free-floating conclusions.

6.7 Search and Indexing Layer

Responsible for:

- Fast text search.
- Entity search.
- Date/time filtering.
- Conversation filtering.
- Fuzzy search for nicknames and aliases.
- Full-text indexing.
- Vector search for semantic similarity if enabled.

6.8 Persistence Layer

Responsible for:

- Local database storage.
- Case files.
- Import metadata.
- Audit logs.
- Review status.
- AI outputs.
- Report configurations.

Recommended local database options:

- SQLite for MVP.
- DuckDB for analytical workloads.
- LiteDB if using .NET document storage.
- PostgreSQL for future team/server deployment.

6.9 Reporting and Export Layer

Responsible for:

- PDF reports.
- Word-compatible reports.
- CSV exports.
- Timeline exports.
- Lead packets.
- Warrant-support summaries.
- Discovery-friendly source reference tables.

6.10 Security and Audit Layer

Responsible for:

- Case-level access controls.
 - Evidence file hashing.
 - Read-only preservation of original imports.
 - Audit logging.
 - AI usage logging.
 - User action logs.
 - Export logs.
 - Optional encryption at rest.
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7. Data Model Overview

7.1 Case

Represents an investigation workspace.

Fields:

- Case ID
- Case number
- Title
- Incident type
- Incident date/time
- Incident location
- Lead investigator
- Agency
- Notes
- Created date
- Last modified date

7.2 Source Import

Represents an imported file or evidence package.

Fields:

- Source ID
- Case ID
- Source name
- Source type
- Owner/person associated
- Device/account associated
- Platform
- Original filename
- File hash
- Import date
- Imported by
- Extraction type
- Provider return type
- Notes

Source types:

- Phone extraction
- Social media return
- Call detail record

- Screenshot package
- Manual entry
- Transcript
- Other

7.3 Device

Represents a physical or logical device.

Fields:

- Device ID
- Owner person ID
- Device label
- Make/model
- Phone number
- IMEI/MEID if available
- Extraction type
- Notes

7.4 Platform Account

Represents a social media or messaging account.

Fields:

- Account ID
- Person ID
- Platform
- Username/handle
- Account numeric ID if known
- Linked phone
- Linked email
- Confidence
- Supporting source references

7.5 Person

Represents a person or unknown subject.

Fields:

- Person ID
- Display name
- Legal name
- Role tags

- Known phone numbers
- Known handles
- Known aliases
- Notes
- Confidence level

Role tags should be user-controlled and carefully worded:

- Victim
- Suspect
- Witness
- Associate
- Unknown
- Investigator-defined

7.6 Identity / Alias

Represents a nickname, handle, phone number, or identity reference that may map to a person.

Fields:

- Identity ID
- Identity type
- Raw value
- Normalized value
- Linked person ID
- Confidence
- Source references
- Investigator review status

Identity types:

- Phone number
- Contact name
- Instagram handle
- Facebook name
- Snapchat username
- TikTok username
- Nickname
- Rap name
- Street name
- Email address
- Unknown account ID

7.7 Message

Represents a normalized message from any platform.

Fields:

- Message ID
- Case ID
- Source ID
- Platform
- Conversation ID
- Thread/source thread ID
- Timestamp original
- Timestamp normalized
- Timezone
- Sender identity
- Recipient identities
- Direction
- Message body
- Body hash
- Attachment references
- Deleted/recovered status
- Read/unread status if available
- Source row/page reference
- Import confidence
- Original metadata JSON

Deleted/recovered status examples:

- Present
- Deleted artifact
- Recovered deleted
- Missing counterpart
- Provider-only
- Screenshot-only
- Unknown

7.8 Call Record

Represents call logs or CDR records.

Fields:

- Call ID
- Case ID
- Source ID
- Timestamp
- Duration
- Caller
- Callee
- Direction

- Call type
- Platform/carrier
- Cell/site info if imported
- Notes

7.9 Attachment

Represents attached files/media.

Fields:

- Attachment ID
- Source ID
- Linked message ID
- Filename
- File hash
- MIME type
- Size
- Metadata
- Extracted text if available
- Thumbnail path if available

7.10 Conversation

Represents a reconstructed thread across one or more sources.

Fields:

- Conversation ID
- Case ID
- Conversation title
- Platform
- Participants
- Source IDs
- Start time
- End time
- Message count
- Reconciliation status
- Review status

7.11 Reconciled Message Group

Represents multiple source records believed to be the same message.

Fields:

- Reconciled Group ID
- Canonical message ID
- Source message IDs
- Match confidence
- Match method
- Investigator review status
- Notes

Match methods:

- Exact body + timestamp + sender/recipient
- Body hash + participant match
- Attachment hash match
- Provider message ID match
- Manual match
- Fuzzy match

7.12 Timeline Event

Represents an item on the unified timeline.

Fields:

- Event ID
- Case ID
- Event type
- Timestamp
- Title
- Description
- Linked messages
- Linked calls
- Linked people
- Linked accounts
- Linked locations
- Source references
- Created by AI/user/system
- Review status

Event types:

- Message
- Call
- Platform switch
- Possible deletion gap
- Key phrase

- Location mention
- Manual case event
- Incident event
- Warrant lead

7.13 Finding

Represents an AI or investigator finding.

Fields:

- Finding ID
- Case ID
- Finding type
- Title
- Summary
- Confidence
- Supporting artifacts
- Contradicting artifacts
- AI-generated flag
- Investigator status
- Created date
- Reviewed by
- Review notes

Finding types:

- Missing message
- Communication spike
- Platform switch
- Unknown number
- Alias link
- Relevant thread
- Location mention
- Weapon/drug/money term
- Warrant lead
- Timeline event candidate

7.14 Lead

Represents an actionable investigative lead.

Fields:

- Lead ID
- Case ID
- Title

- Priority
- Lead type
- Reason
- Supporting findings
- Suggested action
- Assigned to
- Status
- Due date
- Notes

Lead types:

- Identify unknown phone number
- Warrant social media account
- Warrant phone records
- Review thread
- Resolve alias
- Compare source
- Interview subject
- Verify location

7.15 Audit Log

Represents user/system actions.

Fields:

- Audit ID
- Case ID
- User
- Action type
- Entity type
- Entity ID
- Old value
- New value
- Timestamp
- Reason/notes

8. Ingestion Design

8.1 Import Wizard Goals

The import process must be simple enough for a detective to use without needing a forensic analyst every time.

The wizard should:

1. Ask what type of source is being imported.
2. Let the user assign the source to a person, device, or account.
3. Preview the data.
4. Map columns if needed.
5. Validate timestamps and required fields.
6. Show import warnings.
7. Preserve and hash the original file.
8. Import normalized records.
9. Create a clear import summary.

8.2 Supported MVP Inputs

Phase 1 should support:

- CSV message exports.
- XLSX message exports.
- CSV call logs.
- Manual social media message exports in table format.
- Manual message entry.
- Custom column mapping.

8.3 Future Inputs

Future versions may support:

- Celebrite exports.
- Magnet AXIOM exports.
- GrayKey-style exports.
- Oxygen exports.
- XRY exports.
- JSON social media provider returns.
- HTML social media downloads.
- XML exports.
- PDF transcripts where structured parsing is possible.
- Screenshot OCR as a separate, carefully labeled workflow.

8.4 Column Mapping

The import wizard should recognize common field names:

DumpLens Field	Possible Source Columns
Timestamp	date, time, datetime, sent_at, created_at
Sender	from, sender, author, source, account_from

DumpLens Field	Possible Source Columns
Recipient	to, recipient, destination, account_to
Message Body	body, text, message, content
Platform	app, service, platform, source_app
Direction	incoming, outgoing, direction
Thread ID	conversation_id, chat_id, thread, room
Message ID	message_id, id, guid
Attachment	attachment, media, filename

8.5 Import Validation

The import wizard should flag:

- Missing timestamp.
- Unparseable timestamp.
- Missing sender/recipient.
- Empty body with attachment.
- Duplicate rows.
- Unknown timezone.
- Conflicting owner information.
- Inconsistent phone number formats.

Warnings should not always block import. The user should be able to import imperfect data while clearly seeing what needs review.

9. Reconciliation Design

9.1 Purpose

The reconciliation engine compares messages and calls across sources to determine whether they represent the same communication, a related communication, or a missing counterpart.

9.2 Matching Signals

The engine should use weighted signals:

- Platform match.
- Thread ID match.
- Provider message ID match.
- Timestamp proximity.

- Sender/recipient identity match.
- Message body exact match.
- Message body normalized match.
- Message body hash.
- Attachment hash.
- Directional inverse match across devices.
- Conversation participant overlap.
- Manual investigator link.

9.3 Match Statuses

Each message can have reconciliation statuses:

Status	Meaning
Matched	Same message appears in multiple sources.
Source-only	Message appears in only one known source.
Missing counterpart	Message should likely appear in another source but does not.
Recovered deleted	Message was recovered as a deleted artifact.
Provider-only	Message appears in provider return but not device dump.
Screenshot-only	Message appears only in screenshot/manual import.
Ambiguous match	Possible match exists but needs review.
Manual match	Investigator manually linked records.
Excluded	Investigator determined records should not be matched.

9.4 Missing Counterpart Detection

DumpLens should not assume every source-only message means deletion. Instead, it should use careful language.

Example language:

This message appears in the victim phone export but was not located in the suspect phone export. This may indicate deletion, incomplete extraction, platform sync differences, retention differences, or import limitations. Investigator review is required.

9.5 Deletion Gap Detection

A deletion gap may be suggested when:

- A thread is matched before and after a time window.

- One source contains messages inside that window.
- Another source lacks those messages.
- The missing source otherwise contains the thread.
- The window overlaps with the incident or other case-relevant period.

The system should label this as:

- Possible deletion gap
- Possible missing counterpart window
- Needs review

Not:

- Confirmed deletion
- Evidence tampering
- Proof of consciousness of guilt

9.6 Manual Review Workflow

The user should be able to:

- Confirm match.
- Reject match.
- Mark as ambiguous.
- Mark as possible deletion.
- Add review notes.
- Pin to timeline.
- Create a lead.

10. AI / LLM Design

10.1 AI Role

The AI layer should act as a review assistant. It should help the investigator find and organize information. It should never make final legal conclusions.

10.2 AI Modes

Quick Triage Mode

Scans imported data for:

- Threat language.
- Weapon terms.
- Drug terms.

- Money terms.
- Vehicle references.
- Location references.
- Alias/nickname references.
- Deletion/cleanup language.
- Platform switching.
- Unusual communication spikes.

Thread Summary Mode

Summarizes individual conversations with source-backed message references.

Case-Aware Mode

Uses case profile fields to prioritize relevance:

- Incident date/time.
- Incident location.
- Victim names.
- Suspect names.
- Known nicknames.
- Known phone numbers.
- Known vehicles.
- Known gang/group references.
- Investigator keywords.

Timeline Suggestion Mode

Suggests timeline events based on messages, calls, gaps, and case events.

Gap Analysis Mode

Identifies possible missing messages, platform gaps, unknown contacts, unlinked accounts, or suspicious app-switching behavior.

Warrant Lead Mode

Suggests possible warrant targets with source-backed reasons.

10.3 AI Output Requirements

Every AI output must include:

- Finding title.
- Plain-language summary.
- Confidence level.
- Source artifact references.

- Exact messages or records supporting the finding.
- Why the finding matters.
- Limitations or alternative explanations.
- Recommended human review action.

10.4 AI Confidence Levels

Recommended labels:

Confidence	Meaning
High	Strong source support with clear context.
Medium	Relevant support exists but context needs review.
Low	Weak signal or slang/context-dependent.
Unknown	Insufficient context to assess.
Investigator confirmed	Human reviewer confirmed after source review.
Investigator rejected	Human reviewer rejected the AI finding.

10.5 Local LLM Support

DumpLens should support a pluggable LLM framework:

- Local model packs.
- ONNX-compatible default option if feasible.
- Optional local server provider.
- Optional approved cloud provider.
- No-LLM mode for keyword and rules-only analysis.

The system should allow agencies to disable cloud processing entirely.

10.6 Redaction for Cloud AI

If cloud AI is used, DumpLens should support optional redaction before sending text:

- Names replaced with Person A/B.
- Phone numbers replaced with Phone 1/2.
- Addresses replaced with Location 1/2.
- Case numbers removed.
- Exact identifiers kept locally.

The system must log any AI request and response if cloud AI is used.

11. Search and Analysis Features

11.1 Search Bar

The main search should support:

- Exact phrase search.
- Fuzzy search.
- Boolean terms.
- Date range filtering.
- Sender/recipient filtering.
- Platform filtering.
- Source filtering.
- Deleted/recovered status filtering.
- Attachment filtering.
- Reviewed/unreviewed filtering.

11.2 Common Search Shortcuts

The UI should include one-click investigative searches:

- Incident window.
- Before incident.
- After incident.
- Unknown numbers.
- Deleted/recovered messages.
- Missing counterparts.
- Platform switches.
- Weapon terms.
- Vehicle terms.
- Location terms.
- Money terms.
- Cleanup language.
- Threats.
- New contacts near incident.

11.3 Saved Searches

Users should be able to save searches such as:

- "24 hours before homicide."
- "Messages mentioning victim nickname."
- "Unknown numbers after incident."
- "Possible cleanup language."

12. UI / UX Design

12.1 UX Goals

DumpLens should feel:

- Fast.
- Clear.
- Evidence-first.
- Comfortable for non-technical investigators.
- Powerful for analysts.
- Defensible for court/prosecution review.

The app should avoid overwhelming the user with raw tables first. It should guide them through the investigation.

12.2 Core Navigation

Recommended main navigation:

1. **Dashboard**
2. **Sources**
3. **Conversations**
4. **Timeline**
5. **Gaps & Deletions**
6. **Entities & Aliases**
7. **Leads**
8. **AI Findings**
9. **Reports**
10. **Settings**

12.3 Overall Layout Pattern

Most screens should use a three-panel layout:

Top Bar: Case Name Global Search Filters Export		
Left Panel Navigation / Filters	Main Workspace Timeline, conversation, graph, table, review list	Right Panel Details, source refs, actions

This gives the user:

- Left: what they are looking at.
- Center: the work area.
- Right: why it matters and what to do next.

12.4 Top Bar

The top bar should include:

- Case title.
- Case number.
- Global search.
- Current date range filter.
- Source filter.
- Import button.
- Export/report button.
- User/profile menu.

12.5 Dashboard Screen

Purpose: Give the investigator a quick status overview.

Recommended dashboard cards:

- Sources imported.
- Total messages.
- Total calls.
- Conversations reconstructed.
- Possible missing-message gaps.
- High-priority AI findings.
- Open leads.
- Unknown numbers/accounts.
- Incident-window activity.

Main dashboard sections:

1. Case profile summary.
2. Import/source health.
3. Priority findings.
4. Incident-window timeline preview.
5. Open leads.
6. Suggested next review actions.

Example layout:

[Case Summary] [Sources Imported] [Messages/Calls] [Open Leads]

[Incident Window Activity Timeline Preview]

[Priority Findings]

- Possible deletion gap in Suspect ↔ Victim thread
- Unknown 803 number first appears 14 minutes after incident
- Instagram conversation referenced but not recovered from suspect phone

[Suggested Next Review]

- Review missing-message gap from 7:16 PM to 7:45 PM
- Resolve unknown 803 number
- Review co-defendant Instagram DMs

12.6 Sources Screen

Purpose: Manage imports and evidence packages.

Features:

- List all imported sources.
- Show source owner, platform, file hash, import date, record counts.
- Show import warnings.
- Open source details.
- Re-run normalization if mapping changes.
- Add notes about extraction type.

Columns:

- Source name
- Owner/person
- Type
- Platform
- Records imported
- Warnings
- Hash
- Imported by
- Imported date

Source detail panel:

- Original filename.
- File hash.
- Import settings.
- Field mapping.
- Warning list.

- Linked person/device/account.
- Audit history.

12.7 Import Wizard UI

Step-by-step flow:

1. **Choose source type**

2. Phone extraction
3. Social media return
4. Call log
5. Manual spreadsheet
6. Screenshot/manual entry

7. **Assign source**

8. Person
9. Device
10. Platform account
11. Unknown source

12. **Preview data**

13. Show first 50 rows.
14. Detect likely columns.
15. Let user correct mappings.

16. **Confirm timestamp/timezone**

17. Show detected format.
18. Ask user to verify timezone.

19. **Validation results**

20. Warnings.
21. Duplicates.
22. Missing fields.

23. Import summary

24. Records imported.

25. Records skipped.

26. Warnings.

27. Created conversations.

12.8 Conversations Screen

Purpose: Read and analyze reconstructed conversations.

Layout:

Left Panel:

- Conversation list
- Filters
- Participants
- Platforms
- Sources

Center:

- Message thread
- Matched/missing markers
- Attachments
- Calls if included

Right Panel:

- Conversation summary
- Source comparison
- Missing messages
- AI findings
- Actions

Conversation list should show:

- Participants.
- Platform.
- Message count.
- Source count.
- Date range.
- Missing/gap indicator.
- Priority indicator.

Message thread should show:

- Timestamp.
- Sender.
- Body.
- Platform.
- Source icons.
- Matched-source badges.
- Deleted/recovered badges.
- Pin-to-timeline button.
- Source-reference button.

Example source badges:

- Suspect phone
- Victim phone
- Instagram return
- Provider return
- Screenshot

12.9 Source Comparison View

Inside a conversation, the user should be able to switch from normal thread view to side-by-side comparison.

Example:

Suspect Phone	Victim Phone	Status
7:14 PM where you at	7:14 PM where you at	Matched
	7:16 PM Elm St	Missing left
7:19 PM bet	7:19 PM bet	Matched
	7:45 PM why you do that	Missing left

Actions:

- Confirm missing counterpart.
- Mark as extraction limitation.
- Mark as possible deletion gap.
- Create finding.
- Pin to timeline.
- Add note.

12.10 Timeline Screen

Purpose: Build one unified timeline across all platforms and sources.

Timeline should support:

- Chronological view.
- Density view.
- Incident window focus.
- Platform filtering.
- Person filtering.
- Source filtering.
- Message/call/manual event filtering.
- Gaps/deletion markers.
- AI-suggested events.
- Investigator-pinned events.

Timeline item types:

- SMS/iMessage.
- Social media message.
- Call.
- Missed call.
- Attachment.
- Location mention.
- Manual event.
- Possible deletion gap.
- Platform switch.
- Warrant lead.

Timeline visual cues:

- Solid item: source artifact.
- Dashed item: AI-suggested event.
- Warning marker: possible gap.
- Checkmark: investigator-confirmed.
- Question mark: needs review.

12.11 Gaps & Deletions Screen

Purpose: Provide a dedicated review space for missing messages, source-only messages, and possible deletion windows.

Main table columns:

- Gap title.
- Conversation.

- Time window.
- Missing from source.
- Present in source.
- Message count.
- Incident proximity.
- Confidence.
- Review status.

Detail view:

- Side-by-side source comparison.
- Messages before/after gap.
- Messages present only in one source.
- Alternative explanations.
- Suggested review steps.
- Action buttons.

Action buttons:

- Confirm as relevant gap.
- Mark as likely extraction limitation.
- Mark as irrelevant.
- Create lead.
- Pin to timeline.
- Export gap packet.

12.12 Entities & Aliases Screen

Purpose: Resolve people, phone numbers, handles, nicknames, and unknown accounts.

Left panel:

- People.
- Phone numbers.
- Handles.
- Nicknames.
- Unknown identities.

Center:

- Identity graph/table.
- Links between identities.
- Confidence levels.

Right panel:

- Supporting source references.
- AI suggestions.

- Investigator notes.
- Merge/split controls.

Features:

- Merge identities.
- Split incorrect merges.
- Confirm alias.
- Reject alias.
- Add source support.
- Create lead for unresolved identity.

12.13 Leads Screen

Purpose: Manage investigative leads produced by the user, AI, or system.

Lead categories:

- Warrant target.
- Unknown number/account.
- Missing message review.
- Platform gap.
- Alias resolution.
- Interview follow-up.
- Source comparison.
- Report preparation.

Lead fields:

- Title.
- Priority.
- Assigned to.
- Status.
- Reason.
- Supporting artifacts.
- Suggested action.
- Due date.
- Notes.

Statuses:

- Open.
- In review.
- Completed.
- Rejected.
- Deferred.

12.14 AI Findings Screen

Purpose: Review AI-generated findings in a controlled queue.

Finding cards should show:

- Finding title.
- Confidence.
- Finding type.
- Short summary.
- Source count.
- Review status.

The detail panel should show:

- Full AI explanation.
- Supporting artifacts.
- Alternative explanations.
- Investigator actions.

Actions:

- Approve.
- Reject.
- Edit summary.
- Convert to lead.
- Pin to timeline.
- Add to report.

12.15 Reports Screen

Purpose: Create clean outputs.

Report types:

- Case communication summary.
- Incident-window timeline.
- Missing-message gap report.
- Warrant lead packet.
- Unknown identity report.
- AI findings review log.
- Source import manifest.
- Full evidence citation appendix.

Report builder flow:

1. Choose report type.

2. Select date range.
3. Select people/accounts/sources.
4. Choose findings/timeline events.
5. Preview report.
6. Export PDF/DOCX/CSV.

Reports should clearly label:

- Source artifact.
 - AI-assisted summary.
 - Investigator-confirmed conclusion.
 - Unreviewed suggestion.
-

13. User Friendliness Requirements

13.1 Plain-Language Labels

Avoid overly technical wording where possible.

Use:

- “Possible missing message” instead of “asymmetric artifact absence.”
- “Source comparison” instead of “multi-source reconciliation matrix.”
- “Needs review” instead of “unvalidated inference state.”

13.2 Guided Workflows

The app should guide users with next-step prompts:

- “Import another source to compare this conversation.”
- “This thread has messages missing from one source. Review side-by-side?”
- “This unknown number appears near the incident window. Create lead?”
- “This AI finding has not been reviewed.”

13.3 Progressive Disclosure

Default screens should be simple. Advanced details should be available but not forced.

For example:

- Show a simple thread summary first.
- Let the user expand to source hashes, raw metadata, body hashes, and match scoring.

13.4 Review-First Interaction

Important findings should never silently enter reports. The user should review and approve them first.

13.5 Keyboard and Power-User Support

Investigators and analysts should be able to work fast:

- Ctrl+F global search.
- Keyboard shortcuts for next/previous finding.
- Quick pin to timeline.
- Quick mark reviewed.
- Saved filters.
- Bulk select messages.

13.6 Accessibility

UI should support:

- High-contrast mode.
- Resizable text.
- Keyboard navigation.
- Clear icons plus text labels.
- Avoid color-only meaning.

14. Suggested Screen Layouts

14.1 Dashboard Layout

DumpLens Case: 26-04122 Homicide Search... Import	
[Sources: 6] [Messages: 84,233] [Gaps: 12] [Open Leads: 9]	
Incident Window Timeline	Priority Findings
18:58 SMS	- Possible deletion gap
19:03 Call	- Unknown number after event
19:16 Missing counterpart	- IG messages missing
Suggested Next Actions	

[Review Gap] [Resolve Unknown Number] [Open IG Thread]
--

14.2 Conversation Layout

Conversations	Thread: Suspect ↔ Victim	Details
Search/filter	7:14 PM Suspect: where you at	Sources
- High prio	[Suspect Phone][Victim Phone]	Summary
- Missing msg		Gaps
- Instagram	7:16 PM Victim: Elm St	Actions
	[Victim Phone Only] ▲	
		[Pin] [Lead]

14.3 Gap Review Layout

Gap Review: Suspect phone missing messages in matched thread	
Suspect Phone	Victim Phone
7:14 where you at	7:14 where you at
	7:16 Elm St ▲ Missing left
7:19 bet	7:19 bet
	7:45 why you do that ▲ Missing
Explanation: Messages appear in victim source but were not found in suspect source. Review extraction limitations before drawing conclusions.	
[Confirm Relevant Gap] [Extraction Limitation] [Irrelevant]	

14.4 Lead Packet Layout

Lead: Warrant Instagram account @example
Reason
- Suspect told contact to move conversation to Instagram.
- Co-defendant dump contains IG DMs during offense window.
- Suspect dump contains no matching IG messages.

Supporting Artifacts

[Message 1] [Message 2] [Source comparison] [Timeline item]

Suggested Records

- Subscriber/account information
- Direct messages
- Login/IP history
- Linked phone/email

[Approve Lead] [Edit] [Reject] [Export Packet]

15. Reporting Design

15.1 Report Principles

Reports should be clean, readable, and defensible.

Each report should include:

- Case information.
- Source list.
- Method summary.
- Findings.
- Supporting artifacts.
- Limitations.
- Reviewer information.
- Appendix with source references.

15.2 Missing Message Gap Report

Sections:

1. Case summary.
2. Source files compared.
3. Conversation participants.
4. Gap summary.
5. Side-by-side message table.
6. Alternative explanations.
7. Investigator notes.
8. Source reference appendix.

15.3 Warrant Lead Packet

Sections:

1. Proposed target.
2. Reason for review.
3. Supporting artifacts.
4. Related timeline events.
5. Known identifiers.
6. Suggested record categories.
7. Limitations.
8. Investigator approval status.

The report should not state that probable cause exists unless the investigator writes that conclusion separately.

16. Security, Privacy, and Chain-of-Custody Design

16.1 Original Evidence Preservation

Original files should be:

- Copied into a protected evidence store or referenced read-only.
- Hashed on import.
- Never modified.
- Linked to normalized records.

16.2 Audit Logging

Audit log should capture:

- Imports.
- Deletes/removals.
- Field edits.
- AI runs.
- Findings approved/rejected.
- Reports exported.
- Leads created/completed.
- Identity merges/splits.

16.3 User Permissions

Possible roles:

- Admin
- Investigator
- Analyst
- Reviewer
- Read-only viewer

Permissions should control:

- Importing data.
- Running AI.
- Approving findings.
- Exporting reports.
- Editing identities.
- Managing users.

16.4 Encryption

Preferred:

- Encrypt case database at rest.
- Encrypt evidence file store.
- Support case password or agency-managed key.

16.5 Cloud AI Controls

If enabled:

- Require admin approval.
- Log all requests.
- Support redaction.
- Allow per-case disablement.
- Clearly show when a finding used cloud AI.

17. Guardrails and Legal Defensibility

17.1 Required Guardrails

- AI findings must cite source artifacts.
- AI cannot finalize findings without human review.
- Reports must label AI-assisted content.

- Original data must stay read-only.
- Every inference must show confidence and limitations.
- Users must be able to reject or correct AI output.
- Suggested warrants must be framed as investigative leads.
- Unknown identities must remain unknown until supported.

17.2 Fact vs Inference Labels

Every major item should be labeled as one of:

- Source artifact
- System-derived match
- AI-suggested finding
- Investigator-confirmed finding
- Investigator note
- Rejected finding

17.3 Alternative Explanation Prompts

For deletion gaps, the app should remind users of alternatives:

- Incomplete extraction.
- App sync differences.
- Different retention behavior.
- Timezone conversion issue.
- Duplicate account/device confusion.
- Provider export limitations.
- Import mapping error.

18. MVP Definition

18.1 MVP Name

DumpLens MVP: Multi-Dump Message Reconciler

18.2 MVP Scope

The MVP should include:

1. Case creation.
2. CSV/XLSX message import.
3. CSV/XLSX call log import.
4. Source manager.
5. Basic people/identity management.

6. Conversation threading.
7. Cross-source message matching.
8. Missing counterpart detection.
9. Unified timeline.
10. Basic search/filtering.
11. Manual timeline pinning.
12. AI thread summaries.
13. AI entity extraction.
14. AI finding queue.
15. Lead queue.
16. Basic report export.
17. Audit log.

18.3 MVP Exclusions

Do not include in the first MVP:

- Native parsing of every forensic tool format.
- OCR-based screenshot extraction.
- Full graph visualization.
- Team server deployment.
- Advanced permissions.
- Automated warrant drafting.
- Predictive risk scoring.

18.4 MVP Success Criteria

The MVP is successful if an investigator can:

1. Create a case.
2. Import two or more message exports.
3. Assign them to people/devices.
4. View reconstructed conversations.
5. See which messages match across sources.
6. Identify messages present on one source but missing from another.
7. Build a basic incident timeline.
8. Generate a source-backed report.
9. Review AI findings without losing sight of the original evidence.

19. Future Roadmap

Phase 1: MVP

- CSV/XLSX import.

- Case workspace.
- Source manager.
- Conversation view.
- Reconciliation engine.
- Gap review.
- Timeline.
- AI summaries.
- Lead queue.
- Reports.

Phase 2: Better Import Support

- Celebrite/Magnet/GrayKey-style parsers.
- JSON/XML/HTML social media returns.
- Attachment metadata.
- Better call log support.
- Timezone tools.

Phase 3: Advanced Analysis

- Communication spike detection.
- Platform-switching analysis.
- Slang/intel dictionary.
- Alias resolver.
- Entity graph.
- Local semantic search.
- Advanced confidence scoring.

Phase 4: CaseGraph Integration

- Push findings to CaseGraph review queue.
- Link people/incidents/entities.
- Export timeline events to CaseGraph.
- Share evidence-backed claims.

Phase 5: Team/Agency Deployment

- Multi-user case access.
 - Role-based permissions.
 - Agency server mode.
 - Central audit log.
 - Shared dictionaries.
 - Supervisor review workflow.
-

20. Technical Architecture Recommendation

20.1 Recommended MVP Stack

A practical MVP stack could be:

- **Desktop shell:** .NET Avalonia, WPF, or Tauri.
- **Core backend:** .NET or Python service layer.
- **Database:** SQLite for case database.
- **Search:** SQLite FTS5 initially; optional Tantivy/Lucene later.
- **Analytics:** DuckDB optional for large imports.
- **AI:** Pluggable provider interface.
- **Reports:** HTML-to-PDF or DOCX generation library.

20.2 Recommended Architecture Style

Use modular architecture:

```
DumpLens.App  
DumpLens.Core  
DumpLens.Ingestion  
DumpLens.Reconciliation  
DumpLens.Analysis  
DumpLens.AI  
DumpLens.Reporting  
DumpLens.Persistence  
DumpLens.Tests
```

20.3 Provider Interfaces

Important interfaces:

- IMessageImporter
 - ICallLogImporter
 - ISourceHasher
 - IMessageNormalizer
 - IConversationBuilder
 - IMessageReconciler
 - IAiAnalysisProvider
 - IReportExporter
 - IAuditLogger
-

21. Example User Workflow

Scenario: Homicide with Suspect and Victim Phone Dumps

1. Investigator creates case.
 2. Investigator imports victim phone messages.
 3. Investigator imports suspect phone messages.
 4. Investigator assigns each source to a person/device.
 5. DumpLens normalizes messages.
 6. DumpLens detects matching conversations.
 7. Investigator opens the suspect-victim conversation.
 8. DumpLens shows several victim-side messages missing from suspect phone.
 9. Investigator reviews side-by-side comparison.
 10. Investigator marks the gap as relevant but requiring extraction review.
 11. Investigator pins key messages to the timeline.
 12. DumpLens suggests an Instagram warrant lead because the suspect said “hit me on IG” and no IG messages exist in the suspect dump.
 13. Investigator reviews supporting artifacts.
 14. Investigator approves the lead.
 15. Investigator exports a timeline and gap report.
-

22. Open Product Questions

1. Should DumpLens be a standalone product or a CaseGraph module?
 2. Which desktop framework should be used?
 3. What is the first supported import format?
 4. Should AI be local-only at MVP?
 5. Should the MVP include report export or focus only on analysis?
 6. How detailed should the first reconciliation scoring algorithm be?
 7. Should the app support multi-user access from the beginning?
 8. Should evidence files be copied into the case folder or referenced in place?
 9. What exact report formats are needed first?
 10. Should the first design optimize for investigator use or analyst use?
-

23. Recommended Next Step

The next step should be turning this design into a build plan:

1. Define the MVP data schema.
2. Define the import CSV/XLSX format.
3. Create wireframes for the main screens.
4. Break the MVP into development tickets.

5. Build a tiny proof-of-concept with two sample message dumps.
6. Test whether the reconciliation logic correctly identifies matched and missing messages.

The first proof-of-concept should focus on one thing:

Import two message exports from opposite sides of the same conversation and show which messages match, which are source-only, and which are possibly missing from the other device.

That single feature proves the heart of DumpLens.